

Spatial Stories IP2a Testing Plan

Zixu Yang | 48939559 | DECO7230 | Prepared 22 September 2026

Session date _____ Location _____ Tested build identifier _____

Purpose and scope

I will test whether peers can complete the spatial Story creation workflow independently after a standard controller briefing. The study focuses on discoverability, direct manipulation, deletion and publishing in the revised Quest prototype. It will inform the next iteration and Design Evaluation 2.

The prototype uses a Quest 3S with tracked controllers, six floating photo cards, separate text and sticker tool spheres, a spatial bin, and simulated publishing. Barehand tracking, MR passthrough and real Instagram uploading are outside this test. The student reported successful functional checks and casting on 17 September; this is preparation evidence, not participant evaluation.

Evidence from IP1

Desktop IP1 measure	Recorded result
Participants	5
Full completion	4/5
Prompt-free full completion	0/5
Mean ease and spatial naturalness	Approximately 4.8/5 and 4.1/5

Feedback supported retaining floating photos and rotation while improving contextual guidance, sticker visibility, colour meaning, return controls and transition feedback. P02 deletion and publishing were not recorded. Only P01 had a recorded completion time; no IP1 mean time is available.

Research questions

RQ1 Can users discover photo selection, T/S tools and the sequence without task-specific coaching?

RQ2 Can users move, rotate and resize content with controllers, and reach and read the interface comfortably?

RQ3 Do bin feedback, review and confirmation support intentional deletion and publishing?

Course requirement and study choices

The supplied IP2 brief specifies a Week 9 in-person peer test, a prototype, and submission of the testing plan and results as usual. IP2a is not formally graded. This plan chooses five participants and a short session as practical targets; these are not explicit IP2a quotas. Confirm the exact session and submission details in the current course announcement.

Participants and session procedure

Recruit five available adult classmates; keep a sixth form as a spare. Record prior VR, Instagram Story and Spatial Stories experience. Avoid using the developer as a participant. Keep pilots separate from the main dataset and do not quietly replace unsuccessful sessions.

Use moderated task-based observation with brief think-aloud comments and custom post-task questions. Allow approximately five minutes for the core test and questions, plus one to two minutes for fitting, briefing and reset. Optional recovery checks need extra time.

When	Facilitator action
Before timing	Invite voluntary participation; obtain separate permission for recording. Fit headset, explain Trigger, Grip and two-controller resizing consistently. Rehearse controls without demonstrating the core workflow; record any exposure to task actions.
0 to 4 minutes	Read the task, say Begin and start the timer. Observe eight actions. Give help only under the rule below; log the stage and exact wording. Stop timing at visible publication feedback or the four-minute limit.
About 1 minute	Ask ease, naturalness and comfort ratings, the most confusing action, and one suggested change.
After the core test	If time permits, run the optional Back and Undo checks separately. Save notes, reset to Welcome and prepare for the next participant.

Standard task

Choose a photo for a Story and inspect it up close. Add text and a sticker to the photo. Move, rotate and resize at least one editable item. Remove one unwanted item, keeping some content on the Story, then review and publish when you are satisfied.

A short phrase or the available preset text is acceptable; record which was used. Do not require long text entry. Explain that publishing stays inside the prototype.

Help and timing rules

After ten seconds without progress, or an explicit request for help, first ask: “What on the screen might help you continue?” If progress still stops, offer one specific cue and record it. Count each cue as one prompt, including control reminders after timing starts. Record its words and any later action it reveals. Neutral encouragement and repeating the unchanged task are not prompts.

Standard input familiarisation happens before timing and is recorded separately. Independent therefore means independent after this briefing. Keep timing through hesitations and facilitator cues. Stop the core test and timer for discomfort or a blocking technical interruption; mark Interrupted with elapsed time and reason. Any later recovery demonstration stays outside core timing and scores. Record elapsed time for incomplete tasks, not a completion time.

Observation and success definitions

ID	Core action	Successful observation
S01	Choose and inspect a photo	Photo is selected and brought into close editing.
S02	Create and place text	Nonempty text is created and placed on the selected photo.
S03	Create and place a sticker	A sticker is created and placed on the selected photo.
S04	Move an editable item	An item is intentionally moved and released at a chosen position.
S05	Rotate an editable item	An item is intentionally rotated and released.
S06	Resize an editable item	An item changes size using both controllers and remains usable.
S07	Delete an unwanted item	An item is released into the bin and visibly removed.
S08	Review and publish	Participant reviews, confirms publishing and sees completion feedback.

Code each action: I = completed independently; A = completed with help; F = attempted without success; NT = not attempted; NR = not recorded. An earlier hint that supplies an action also makes that later action assisted. Record failures and retries even if the action is eventually completed.

Full completion requires all eight actions scored I or A, including confirmed publication. Partial means at least one action succeeded but the full task did not. Failed means none succeeded. Use Interrupted for a technical or comfort stop and NR if the overall result cannot be established. The app's "complete" log only records publishing and cannot replace this assessment.

Predeclared formative targets

Measure	Target and reporting
Independent full workflow	At least 80% of all core-task starters complete every action with zero task prompts; 4/5 if five start.
Individual core actions	Aim for at least 80% independent success on each observed attempted action; also report NT and NR counts.
Subjective experience	Average ease and naturalness at least 4/5. Explore comfort separately; aim for at least 4/5 and investigate each discomfort report.
Critical issues	No same critical issue in two or more participants. A critical issue blocks required progress or needs a restart or facilitator recovery.

These are design decision thresholds for a small formative study. They do not establish population-level usability, safety or a statistically proven improvement.

Analysis and next design decisions

Use the handwritten or typed observation sheets as the primary record. Optional video and event logs help check uncertain sequences. Link every file to a participant code and, where available, the runtime session ID. Keep raw evidence unchanged and record corrections separately.

Output	Calculation or interpretation
Whole-task results	Report complete, partial, failed, interrupted and unrecorded counts among all starters N. Report confirmed independent completions n/N. Do not count unknown outcomes as known failures; if k could still be independent but cannot be determined because action or prompt records are missing, give the possible interval n/N to (n+k)/N. Known incomplete or assisted cases do not belong in k.
Action results	For each action, attempted observations = I + A + F; independent rate = I / attempted observations. Show NT and NR separately. If there are no observed attempts, report not evaluable.
Prompts and ratings	List each participant's exact prompt count. Calculate each rating mean only from supplied ratings and show its valid n. Zero prompts is a recorded zero; blank is missing.
Time	Report completion times only for completed tasks with recorded timing, including valid n and range or median. Report incomplete elapsed times and interruptions separately. Do not compare with an invented IP1 average.
Findings	Group repeated problems by stage and distinguish observed behaviour, participant comments and interpretation. For each finding, link participant IDs, counterexamples, the proposed change and how it will be retested.

Optional checks and comparison limits

After core measures, ask users to return to the album and revisit their edited photo. For Undo, start a fresh editing opportunity, delete one item and immediately ask for recovery: the current default undo window is about five seconds and applies to the latest deletion. Keep optional success, time and prompts separate; mark unrun checks NT.

IP1 used desktop input. IP2a changes the device, briefing and tasks, and may include repeat participants. Compare patterns cautiously; differences cannot be attributed solely to the redesign. Report prior exposure and technical issues. Do not claim that controller results validate barehand MR gestures.

Evidence handling and AI acknowledgement

Use participant codes rather than names. Explain that participation is voluntary and participants may stop. Obtain recording permission separately; observation can proceed without recording. Keep identifiable footage private and publish only appropriately anonymised results. Stop the session if the participant reports discomfort.

Sources: supplied IP2 Functional Prototype brief; Spatial Stories IP1 Testing Plan; IP1 Key Findings; Design Evaluation 1; current full-concept README and runtime source. ChatGPT/Codex assisted with Unity implementation and troubleshooting and with drafting this protocol, bilingual materials and blank templates. All IP2a results remain uncollected in this pack. See the accompanying AI use statement for scope and actual prompt excerpts.